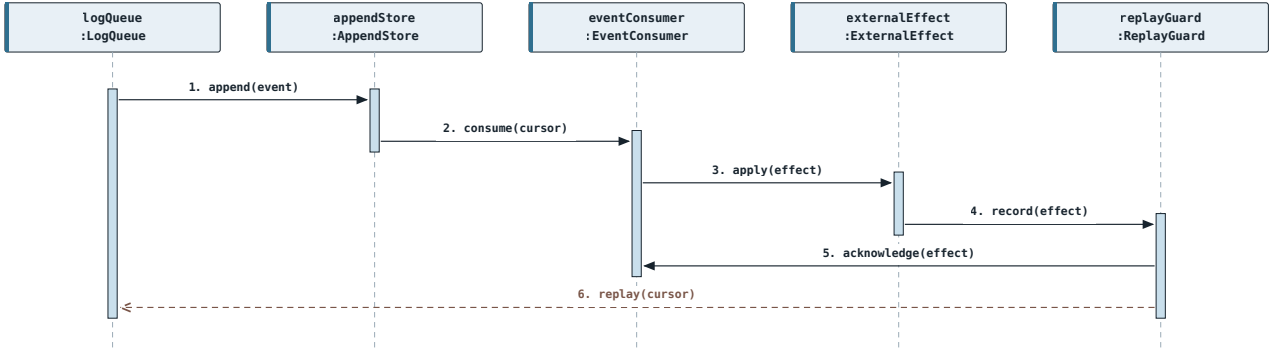
P13 Log-as-queue loops

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Drive work from a durable execution record.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE	
CONTEXT	New work is represented by durable append-only records.
TRIGGER	A consumer polls or subscribes to unseen records.
STRUCTURE	Event identity, append order, consumer cursor, and acknowledgement.
EFFECT	Work advances from the log without replaying already completed effects.

CONTRACT / REQUIRED BEHAVIOR	
REQUIRES	Records have stable identity and consumers retain progress.
ENSURES	Each effectful unit has an idempotency or acknowledgement rule.
INVARIANT	Consumer crash does not erase unacknowledged work.
VIOLATION	Duplicate delivery is deduplicated or documented as at-least-once.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.